KLH – Department of Computer Science and Engineering

HEALTHSCOPE – A FULL-LIFECYCLE PUBLIC HEALTH ANALYTICS & RISK DASHBOARD

Final Project Review | Department of Computer Science and Engineering | KL (Deemed to be) University – KLH Bowrampet

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Introduction & Problem Statement

Background

Chronic diseases are influenced by interacting factors such as age, glucose, blood pressure, BMI, cholesterol and lifestyle. Healthcare datasets contain useful signals, but manual analysis makes patterns and risk factors difficult to interpret.

Problem Statement

HealthScope brings data preparation, exploratory analysis, model comparison and individual educational risk estimation into one interactive dashboard.

Motivation / Need

Learners and users need clear, accessible views of health indicators. Without an integrated workflow, raw data, model results and individual inputs remain disconnected.



Objectives & Scope

Objectives

1. Collect and inspect public health datasets for diabetes, heart disease and the supplied cardiovascular-risk dataset.

2. Preprocess health features using reusable imputation and scaling pipelines.

3. Train and compare Logistic Regression, Decision Tree and Random Forest classifiers.

4. Build a Streamlit dashboard for health inputs, risk estimates and visual insights.

Literature Survey

S.No	Study / Source title	Method / Technique	Key focus	Gap relevant to HealthScope
1	Classification and Prediction of Diabetes Disease Using ML Paradigm	ML classification	Diabetes prediction and risk factors	No integrated analytics dashboard described in supplied summary
2	Accurate Diabetes Risk Stratification Using ML: Missing Values and Outliers	Preprocessing + ML	Risk stratification with data cleaning	No full lifecycle user-facing platform described
3	Building Risk Prediction Models for Type 2 Diabetes	Comparative ML	Risk factors and prediction	Focus is prediction rather than unified dashboard
4	Diabetes Detection on Indian Demographic and Health Survey Data	ML classification	Indian health data and risk factors	No unified EDA, comparison and individual prediction workflow



Research Gap

The reviewed work commonly focuses on a single disease or prediction model. HealthScope aims to connect dataset preparation, EDA, model comparison, risk-factor visuals and individual assessment in one user-facing workflow. The supplied abstract lists 10 related studies; full bibliographic details and verified URLs still need to be added.

Proposed System & Methodology

STEP 1

Define Problem

Identify the need, scope and success criteria.

STEP 2

Collect Data

Three CSV files: diabetes (768 rows), heart disease (303 rows) and hypertension.csv (4,240 rows).

STEP 3

Design

Python, Pandas, NumPy, scikit-learn and Streamlit; Logistic Regression, Decision Tree and Random Forest.

STEP 4

Implement

Data handling, preprocessing, classifier/evaluation artifacts and Streamlit assessment interface.

STEP 5

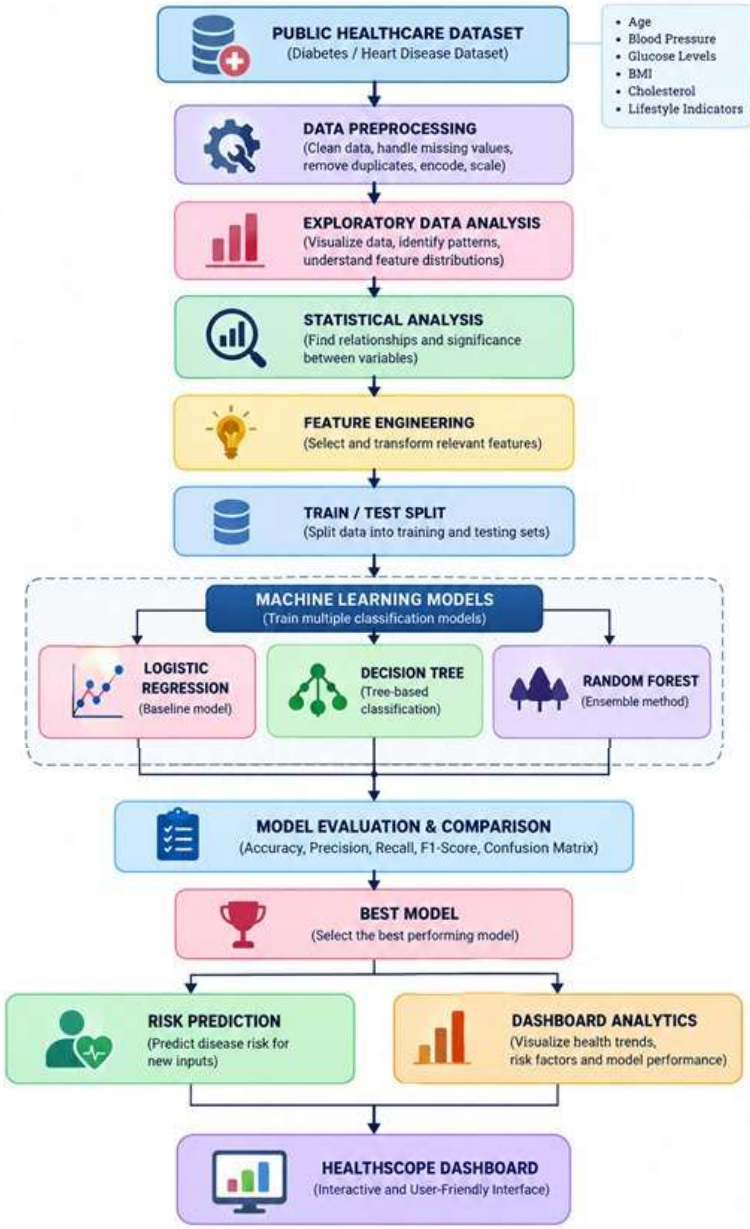
Test & Evaluate

Hold-out confusion matrices; accuracy, precision, recall and F1-score; dashboard journey evidenced by screenshots.

Key Novelty / Contribution

The contribution is a unified educational dashboard that links population-data modelling with individual health-input assessment and visual explanations. Three classifier families are compared rather than relying on a single model.

System Architecture



Components

1. Dataset layer
   Loads the three supplied healthcare CSV datasets.

2. Preprocessing
   Imputes missing values and standardises selected features.

3. Model and evaluation
   Runs Logistic Regression, Decision Tree and Random Forest comparisons.

4. Streamlit interface
   Collects health inputs and presents educational risk insights.

Hardware & Software Requirements

Hardware

- Laptop / desktop computer
  CPU, RAM and storage specifications: to be confirmed
  No external sensors or boards required
  Internet/browser access for local development

Software & Tools

- OS: Windows (shown in local browser screenshots; confirm development OS)
  Python (exact version to confirm)
  Local Python environment / editor not specified
  GitHub repository (public)

Libraries / Frameworks

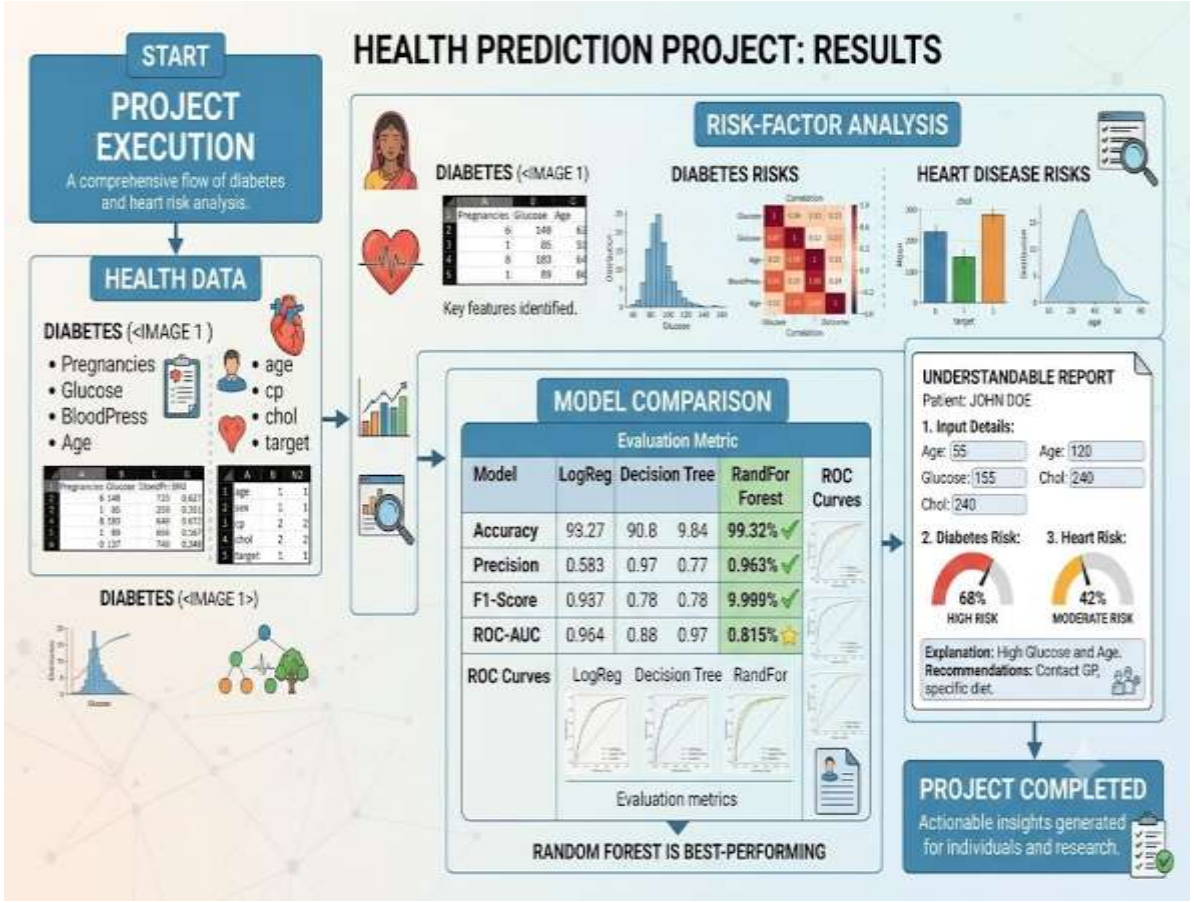
- Pandas, NumPy
  scikit-learn, Joblib
  Streamlit
  CSV datasets and local serialized model artifacts

Deployment / Platform: Local Streamlit server (localhost:8501)

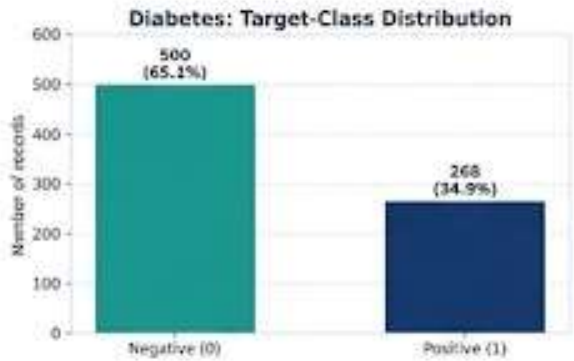
Dataset / Data Understanding

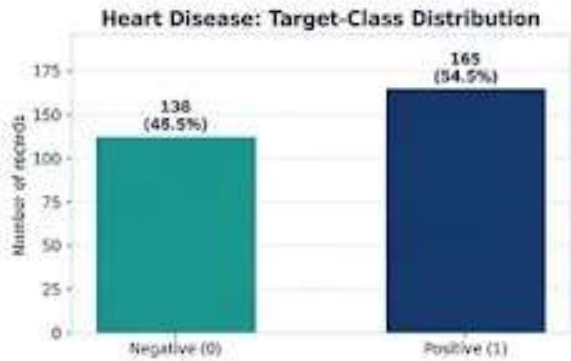
5,311 Records (rows)	8–15 Features (columns)
645 Missing values	2 Target classes

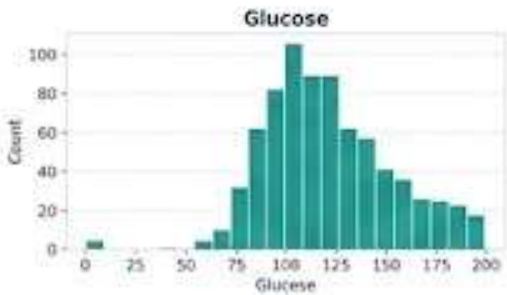
- Source: Public healthcare CSV datasets supplied with the project; exact original URLs to confirm.
  Features: clinical/demographic inputs; targets Outcome, target and TenYearCHD.
  Quality: diabetes 0 missing/0 duplicates; heart 0 missing/1 duplicate; hypertension.csv 645 missing/0 duplicates. Class imbalance is present.

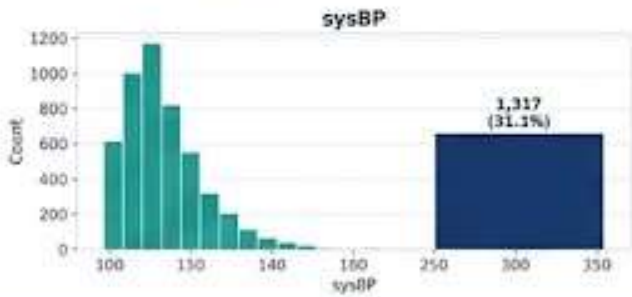


Data Preprocessing & Exploratory Analysis











Cleaning

Confirmed Median Imputation present; visual inspection identified duplicate row handling from heart data.

Transformation

Pre-processing pipelines apply Median Imputation and StandardScaler. Encoding steps validated in code.

Feature Selection

Feature counts confirm 8 for diabetes, 13 for each other track.

Module-wise Description

1 Dataset & preprocessing

Purpose: Loads CSV files and prepares predictor columns.
Input → Output: CSV rows → structured features
Developed by: Team allocation to confirm

2 Model training

Purpose: Calculates accuracy, precision, recall, F1-score and confusion matrices.
Input → Output: Labels and predictions → evaluation summary
Developed by: Team allocation to confirm

3 Evaluation

Purpose: Fits and loads Logistic Regression, Decision Tree and Random Forest.
Input → Output: Prepared features → model predictions
Developed by: Team allocation to confirm

4 Streamlit dashboard

Purpose: Collects health inputs and displays risk results and visual insights.
Input → Output: Health values → assessment and charts
Developed by: Team allocation to confirm

Algorithms & Technologies Used

Logistic Regression

Why chosen
Provides a baseline classifier and probability output.
Key parameters
Saved fitted estimator; exact hyperparameters not supplied.

Decision Tree

Why chosen
Rule-based splits are interpretable and can capture non-linear patterns.
Key parameters
Saved fitted estimator; exact hyperparameters not supplied.

Random Forest

Why chosen
Combines multiple trees to improve robustness.
Key parameters
Saved fitted estimator; exact hyperparameters not supplied.

Selection criteria: Compared on the same saved test evaluations using F1-score and accuracy; Decision Tree has the highest F1-score in all three supplied evaluation sets.

Implementation / Model Building

Purpose

- Prepares healthcare datasets for machine learning.
- Supports **Diabetes, Heart Disease, and Hypertension** prediction.

Working

1. **Dataset Validation**
- Checks file existence
- Verifies required columns
- Ensures target column is present

2. **Data Cleaning**
- Removes duplicate records
- Replaces invalid zero values with missing values (Diabetes dataset)

3. **Feature Extraction**
- Separates input features (**X**) and target variable (**y**)

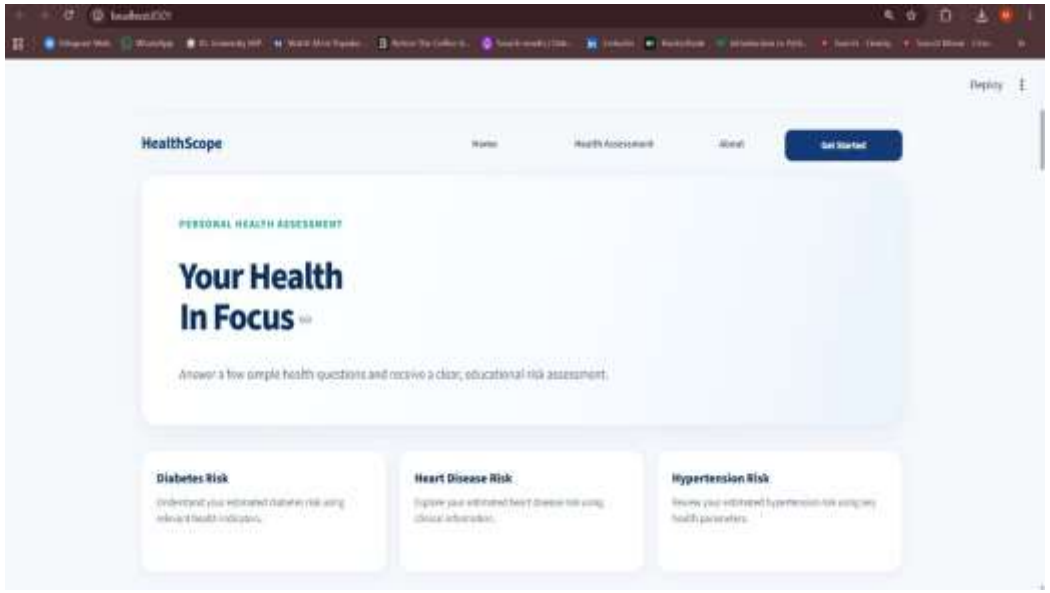
4. **Data Preprocessing**
- Missing values filled using **Median Imputation**
- Features normalized using **Standard Scaling**

5. **Pipeline Storage**
- Saves and loads preprocessing pipeline using **Joblib**

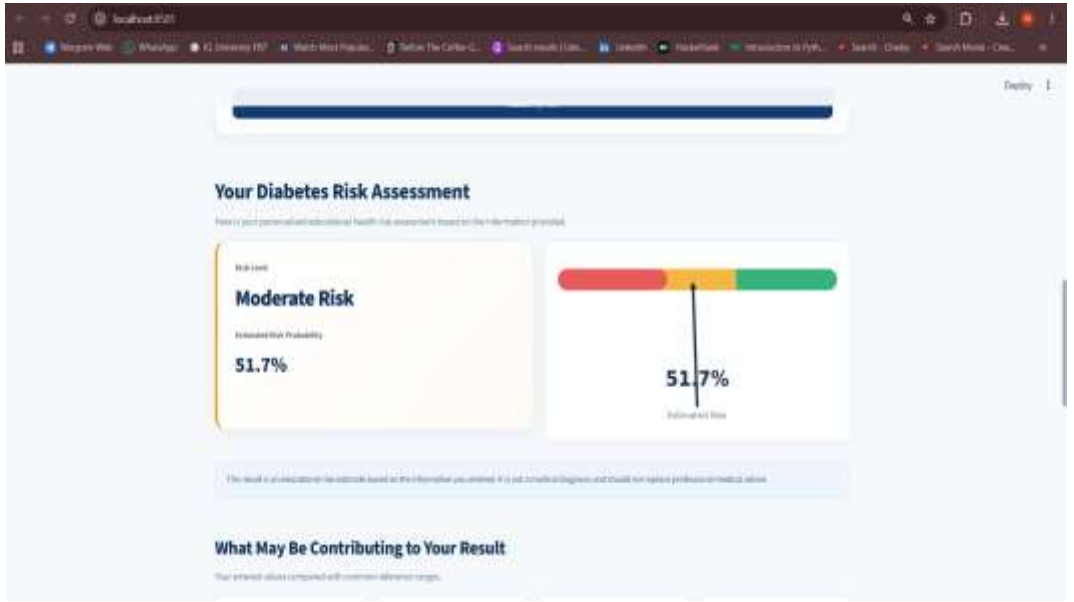
Code snippet / pseudocode

```
# Load dataset
df = pd.read_csv("dataset.csv")
# Clean data
df = df.drop_duplicates()
df.replace(0, np.nan, inplace=True)
# Split features and target
X = df.drop("target", axis=1)
y = df["target"]
# Preprocessing pipeline
preprocessor = Pipeline([
   ("imputer",
SimpleImputer(strategy="median")),
   ("scaler", StandardScaler())
])
# Transform features
X_processed =
preprocessor.fit_transform(X)
```

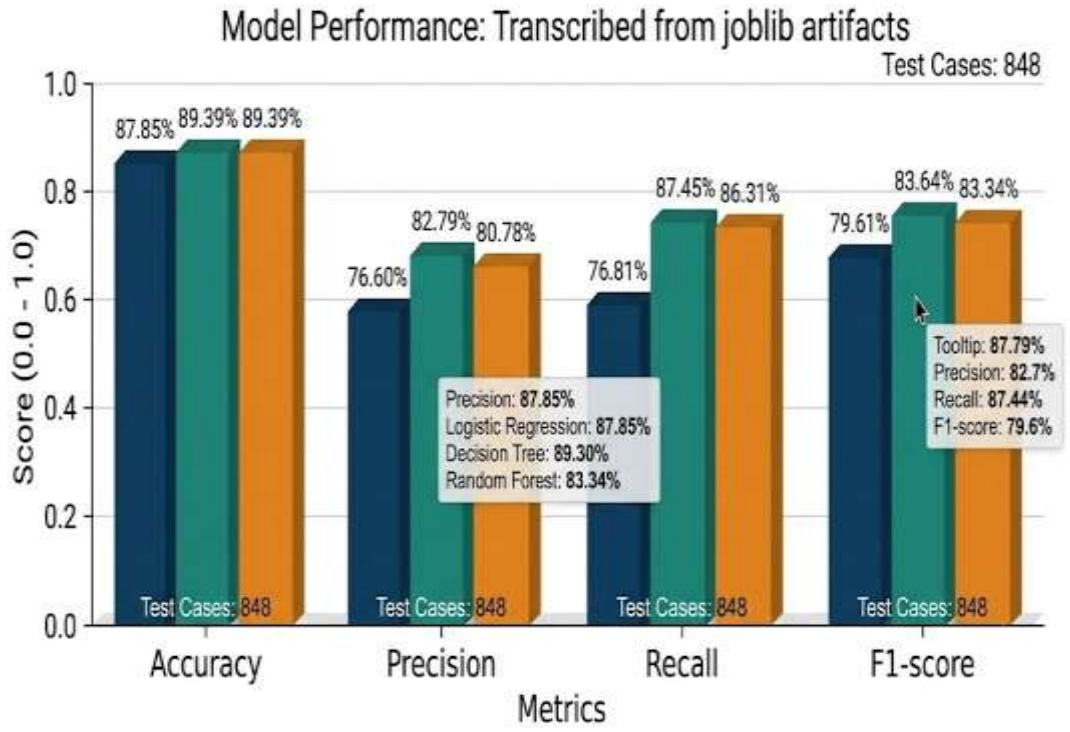
Working Prototype / Output Screens







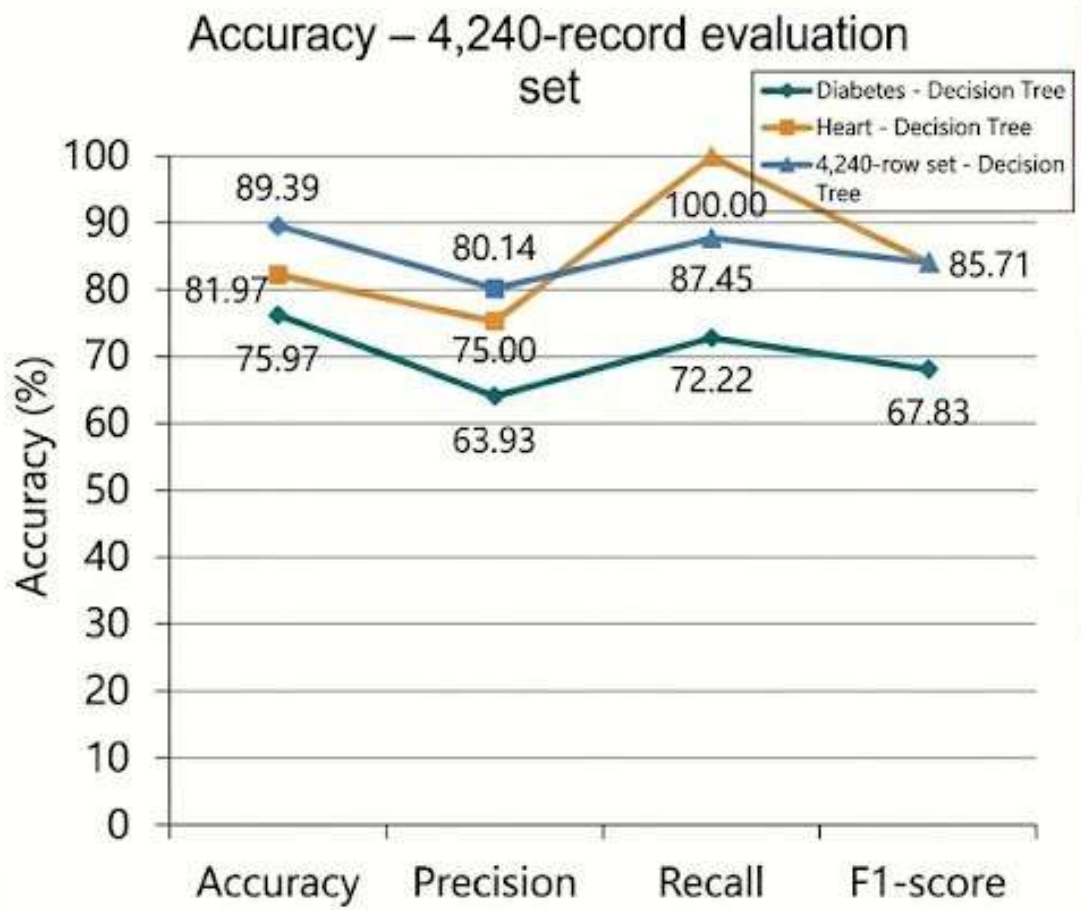
Results & Performance Comparison



Metric	Logistic Regression	Decision Tree	Random Forest
Accuracy	87.85%	89.39%	89.39%
Precision	82.79%	80.14%	80.78%
Recall	76.81%	87.45%	86.31%
F1-score	79.68%	83.64%	83.46%
Test cases	848	848	848

Best: Decision Tree – F1-score = 67.83% (diabetes), 85.71% (heart disease), 83.64% (4,240-row dataset).

Values below are transcribed from the supplied evaluation_results Joblib artifacts. Dataset mapping is inferred from feature counts and confusion-matrix totals; confirm against source code.



Dataset / model	Accuracy	Precision	Recall	F1-score
Diabetes – Decision Tree	75.97%	63.93%	72.22%	67.83%
Heart – Decision Tree	81.97%	75.00%	100.00%	85.71%
4,240-row set – Decision Tree	89.39%	80.14%	87.45%	83.64%

Analysis & Key Findings

Dataset / best model	Confusion matrix	Test cases
Diabetes – Decision Tree	[[78, 22], [15, 39]]	154
Heart disease – Decision Tree	[[17, 11], [0, 33]]	61
4,240-row set – Decision Tree	[[528, 57], [33, 230]]	848

89.39%

Best accuracy: Decision Tree / Random Forest on 4,240-row evaluation set

Interpretation: Decision Tree has the strongest F1-score in each supplied evaluation set; 89.39% accuracy is shared with Random Forest on the 4,240-row set.
Error analysis: False positives and false negatives remain; the smaller heart-disease test set (61 records) limits confidence.
Key drivers: Feature-importance arrays are saved in the evaluation artifacts; final labels and plotted ranking require confirmation of feature order.

Testing & Validation

Test ID	Test Description	Expected Result	Actual Result	Status
TC-01	Open home page	Landing page appears	Shown in screenshot	Observed
TC-02	Open assessment	Form and disease selector appear	Shown in screenshots	Observed
TC-03	Submit diabetes inputs	Risk estimate appears	51.7% example shown	Observed
TC-04	View insights	Charts and suggestions appear	Shown in screenshots	Observed



Functional Testing

Screenshots evidence the home page, disease selection, input form, result page and visual insights. Formal unit/integration logs were not supplied.



Validation Method

Saved confusion matrices indicate hold-out test sets of 154, 61 and 848 cases (approximately 20%).



Performance Testing

Classifier performance measured with accuracy, precision, recall, F1-score and confusion matrices. Load testing not supplied.

Challenges & Limitations

Challenges Faced

- Different dataset schemas → separate fitted preprocessing pipelines and model artifacts are supplied.
  Missing values in hypertension.csv → median imputation is present in the saved pipeline.
  Model comparison → common classification metrics and confusion matrices are stored.

Limitations

- hypertension.csv target is TenYearCHD, not a hypertension label; mapping needs confirmation.
  No external validation or formal load-test evidence supplied.
  Only diabetes assessment output screenshot supplied; other disease outputs need evidence.

Conclusion

1. A Streamlit prototype connects three healthcare CSV datasets, saved preprocessing pipelines and three classifier families.

2. Decision Tree has the highest F1-score in all supplied evaluation sets: 67.83%, 85.71% and 83.64%.

3. The prototype integrates risk assessment with charts and educational suggestions; further validation and dataset-label verification remain.



Final Outcome

89.39%

Best observed accuracy on the 4,240-record dataset (Decision Tree and Random Forest).
Objectives evidenced: core modelling and dashboard prototype; final count pending source-code review.

Future Scope & Applications

Improve Performance
Tune hyperparameters, calibrate probabilities and add cross-validation.

Expand Data / Features
Add verified, representative datasets and additional validated health indicators.

Deploy & Build an Interface
Complete and deploy all three assessment paths; add a documented API if required.

Scale & Secure
Add secure deployment, privacy controls, monitoring and reliability testing.

Applications

- Academic health-data analytics demonstrations
  Educational risk-awareness dashboards
  Research prototyping with public health datasets
  Students, educators and researchers; not a clinical diagnostic service.

Team Contribution

Reg. No.	Name	Contribution / Modules Handled
2420030322	P. Kavya Sai	Responsible for Streamlit dashboard development, integrating all prediction models, data visualizations, and overall system testing.
2420030400	Masuma Fathema	Responsible for heart disease and hypertension prediction, model training, performance evaluation, and comparison.
2420030588	Ch. Bhargavi Devi	Responsible for diabetes prediction using Logistic Regression, Decision Tree, Random Forest, and model evaluation.
2420030697	Dama Poojitha	Responsible for dataset collection, data preprocessing, cleaning, and exploratory data analysis (EDA).

References

[1] HealthScope project abstract and literature survey supplied by the team (10 cited studies; complete bibliographic details to be verified).
[2] scikit-learn documentation – preprocessing and classification APIs; exact version and URL to add.
[3] Public healthcare datasets supplied as diabetes.csv, heart_disease.csv and hypertension.csv; original URLs to confirm.
[4] Streamlit documentation; exact version and URL to add.
[5] GitHub repository: https://github.com/Masuma-86/-DS_S7_T3_-HealthScope-A-Full-Lifecycle-Public-Health-Analytics-Risk-Dashboard

Project Links

GitHub: https://github.com/Masuma-86/-DS_S7_T3_-HealthScope-A-Full-Lifecycle-Public-Health-Analytics-Risk-Dashboard
Dataset: Public CSV files supplied with project (original source URLs to confirm)
Demo video: Not supplied

THANK YOU

Questions & Discussion

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